

UNITED HOLLOW REPUBLIC

OFFICE OF MARITIME AVIATION

H-71 ROTARY-WING AIRCRAFT FAMILY

AH-71 & UH-71

Technical & Operational Familiarization Document

Program: H-71 Rotary-Wing Aircraft Family

Aircraft: AH-71 Attack / Reconnaissance

Aircraft: UH-71 Utility / Assault

Development Partners: Prelancer Technologies / ELTRON Aviation

Issued By: United Hollow Republic Department of Defense

Document Type: Technical & Operational Familiarization

Revision: 1.5

1. INTRODUCTION

The **H-71 aircraft family** represents a new generation of military rotary-wing capability developed for the United Hollow Republic Department of Defense through a joint effort between **Prelancer Technologies** and **ELTRON Aviation**.

The family consists of two principal aircraft:

AH-71 - Attack / Reconnaissance Helicopter

UH-71 - Utility / Assault Helicopter

Both aircraft share a common mechanical and digital architecture while being configured for different operational roles.

The H-71 program was designed around a simple principle:

One aircraft family. Multiple battlefield roles. Common systems.

For pilots, this means an aircraft designed around predictable handling, high control authority, modern flight assistance, and comprehensive aircraft information.

For ground units, it means a helicopter family with recognizable capabilities, common support requirements, and clearly defined methods of interaction with friendly forces.

This document provides a general introduction to the H-71 family and is intended to familiarize personnel with its capabilities and operating characteristics.

2. THE H-71 FAMILY

The H-71 is a medium-weight military helicopter family built around a common airframe and systems philosophy.

Although the AH-71 and UH-71 perform different missions, they share many of the same fundamental systems.

Common systems include:

- twin-engine propulsion;
- four-blade composite main rotor;
- common transmission architecture;
- digital flight-control systems;
- integrated aircraft computers;
- modern navigation equipment;
- digital communications;
- aircraft health monitoring;
- defensive-system interfaces;
- and standardized mission-system interfaces.

This commonality allows the two aircraft to operate alongside one another while reducing the amount of specialized infrastructure required to support them.

3. DESIGN PHILOSOPHY

The H-71 was not designed around a single mission.

Instead, the aircraft was designed as a flexible platform capable of supporting multiple configurations.

The result is a family of aircraft that can be adapted to different operational requirements without fundamentally changing the underlying platform.

The design places particular emphasis on:

CONTROL	High control authority and predictable aircraft response.
POWER	Substantial engine reserve for demanding operating conditions.
INFORMATION	Integrated digital systems that present pilots with relevant aircraft and mission information.
SURVIVABILITY	Redundant systems and integrated defensive capabilities.
MAINTAINABILITY	Continuous aircraft health monitoring and common components.
ADAPTABILITY	Physical and digital capacity for future systems.

4. AH-71

ATTACK / RECONNAISSANCE HELICOPTER

The AH-71 is the dedicated armed reconnaissance and attack configuration of the H-71 family.

Its primary roles include:

- armed reconnaissance;
- battlefield observation;
- close air support;
- escort;
- forward target acquisition;
- precision engagement;
- and tactical overwatch.

The aircraft combines a highly maneuverable flight platform with an integrated sensor and weapons system.

The AH-71 is intended to operate close to the battlefield while maintaining the ability to rapidly transition between observation, engagement, and repositioning.

5. AH-71 SENSOR SYSTEM

The AH-71 incorporates a stabilized multispectral sensor turret.

The system combines several sensor types into a single mission package, including:

- electro-optical imaging;
- infrared imaging;
- low-light imaging;
- laser ranging;
- image stabilization;
- and automated tracking.

The sensor system allows the crew to observe areas of interest without necessarily pointing the entire aircraft directly toward them.

Sensor information is integrated with aircraft navigation and mission data.

This allows the crew to establish a common understanding of:

- aircraft position;
 - sensor position;
 - target location;
 - direction of observation;
 - and selected mission information.
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6. AH-71 WEAPONS SYSTEM

The AH-71 uses a modular external stores architecture.

Depending upon mission configuration, the aircraft can carry combinations of:

- guided air-to-ground weapons;
- unguided rockets;
- air-to-air weapons;
- external fuel;
- reconnaissance equipment;
- and specialized mission stores.

A forward-mounted automatic cannon provides an integrated direct-fire capability.

The aircraft's fire-control system assists the crew by accounting for aircraft attitude, target range, sensor line-of-sight, aircraft movement, and weapon characteristics.

This allows the pilot and mission crew to concentrate on aircraft positioning and target identification rather than manually compensating for every firing variable.

7. AH-71 FLIGHT CHARACTERISTICS

The AH-71 was designed with maneuverability as a primary requirement.

The aircraft maintains substantial control authority during:

- low-speed maneuvering;
- rapid changes in direction;
- confined-area flight;
- terrain-following operations;
- hover transitions;
- and aggressive pitch and roll inputs.

Digital flight-control systems continuously monitor aircraft attitude, rotor loading, engine torque, airspeed, and other parameters.

The system assists the pilot without removing direct control of the aircraft.

For pilots, the result is an aircraft that is intended to remain predictable across a wide range of flight conditions.

8. UH-71

UTILITY / ASSAULT HELICOPTER

The UH-71 is the utility and assault configuration of the H-71 family.
It is designed to provide flexible support to UHR ground and aviation forces.

Primary roles include:

- troop transport;
- combat assault;
- cargo transport;
- medical evacuation;
- search and rescue;
- command and control;
- special operations support;
- and general utility operations.

Where the AH-71 is optimized around observation and engagement, the UH-71 is optimized around **moving people and equipment where they are needed**.

9. UH-71 CABIN

The UH-71 cabin uses a modular internal configuration.
Standardized mounting points allow the aircraft to be rapidly configured for different missions.
Possible configurations include:

PERSONNEL (GUNSHIP)	Troop and passenger transport.
CARGO (LOGISTICS)	Internal equipment and supply transport.
MEDICAL (MEDVAC)	Casualty evacuation and medical support.
COMMAND (CMD)	Additional communications and equipment.
SPECIALIZED, MOUNTABLE	Mission-specific equipment packages.

Large side access doors facilitate rapid loading and unloading.

The cabin floor is reinforced to support both personnel and cargo operations.

10. UH-71 EXTERNAL LOAD CAPABILITY

The UH-71 can carry external loads through a dedicated structural load path.

The aircraft's flight-control and mission computers account for changes in:

- aircraft gross weight;
- center of gravity;
- available engine power;
- and external-load configuration.

This provides the pilot with predictable handling as the aircraft transitions between unloaded, internally loaded, and externally loaded conditions.

For ground units, this capability allows the UH-71 to function as more than a personnel transport.

Maintaining micro flight adjustments to the internal flight computer, allows the pilot to focus on mission, removing loss of control or low-control during changing mission standards.

The aircraft can serve as a flexible aerial logistics platform capable of moving equipment directly to areas that may be difficult to reach by conventional ground transport.

11. PROPULSION

Both H-71 variants use a twin-engine turboshaft configuration.

The propulsion system was selected with reserve power in mind.

This reserve supports operations involving:

- Extreme High and low temperatures;
- high-altitude landing zones;
- heavy aircraft loading;
- rapid climb requirements;
- external loads;
- and single-engine contingency conditions. (in the case of engine failure)

The aircraft continuously monitors engine parameters including:

- torque;
- gas-generator speed;
- turbine temperature;
- fuel flow;
- compressor performance;
- and vibration.

Engine information is presented to the crew through the aircraft's digital monitoring system.

12. ROTOR AND FLIGHT SYSTEM

The H-71 uses a four-blade composite main rotor.

The rotor system was designed to provide a balance between:

- lift;
- maneuverability;
- vibration control;
- high-speed stability;
- and low-speed handling.

The flight-control system monitors aircraft conditions and assists with maintaining the desired response.

Available flight-assistance functions include:

- attitude hold;
- hover stabilization;
- velocity hold;
- automatic trim;
- and power compensation.

These systems are intended to reduce workload during routine flight while preserving direct pilot authority.

13. COCKPIT AND AVIONICS

The H-71 cockpit is built around a digital information architecture.

Multifunction displays provide access to:

- flight information;
- navigation;
- aircraft systems;
- mission information;
- sensor imagery;
- communications;
- and warnings.

Information is prioritized according to operational relevance.

Flight-critical information remains readily available while secondary information can be accessed without requiring dedicated displays for every individual system.

The result is a cockpit designed to provide the crew with a comprehensive picture of the aircraft without requiring constant manual management of individual subsystems.

Current pilots will find familiarity with the 2 by 2 layout, maintaining the central information panel present from previous designs.

14. HELMET-MOUNTED DISPLAY

The H-71 supports helmet-mounted display technology.

The system can present:

- flight references;
- navigation information;
- sensor line-of-sight;
- aircraft warnings;
- target information;
- and selected mission data.

This includes a direct 'Look-through' of the cockpit, allowing the pilot(s) to observe terrain and airspace looking through the visual obstruction of the airframe.

On the AH-71, the system can be integrated with the aircraft's sensor and weapons systems.

This allows the crew to direct attention toward an area of interest while maintaining access to relevant aircraft and mission information.

15. SURVIVABILITY

Survivability is integrated into the H-71 architecture rather than being treated as a separate equipment package.

The aircraft provides interfaces for:

- radar warning;
- missile warning;
- laser warning;
- infrared countermeasures;
- expendable countermeasures;
- electronic warfare systems;
- infrared signature reduction;
- and ballistic protection.

The aircraft computer can combine information from multiple warning systems into a consolidated threat display.

This allows the crew to identify relevant warnings without requiring them to manually interpret multiple independent systems.

16. WHAT GROUND UNITS SHOULD EXPECT

The H-71 family is intended to operate as part of a larger combined-arms environment.

Ground personnel may encounter the aircraft in several different roles.

AH-71

Ground units may see the AH-71 conducting:

- reconnaissance;
- overwatch;
- target observation;
- escort;
- close air support;
- and armed response.

The AH-71 may remain at a distance while its sensors observe an area, or maneuver closer when required by the mission.

UH-71

Ground units may encounter the UH-71 during:

- troop movement;
- casualty evacuation;
- resupply;
- equipment transport;
- assault operations;
- search and rescue;
- or command-support missions.

The UH-71 is intended to operate as a direct extension of ground mobility.

17. GROUND-AIR COORDINATION

The H-71's digital communications and navigation architecture allow it to operate as part of an integrated force.

The aircraft can exchange relevant information with compatible UHR units.

Depending upon mission configuration, this can include:

- aircraft position;
- navigation information;
- sensor information;
- mission information;
- and selected target information.

The objective is to reduce the separation between aviation and ground units.

A ground unit should not have to treat the helicopter as an isolated asset.

The H-71 is designed to function as another connected element of the battlefield network.

18. MAINTENANCE AND SUPPORT

The H-71 incorporates an aircraft health-monitoring system designed to assist maintenance personnel.

Sensors throughout the aircraft monitor the condition of major systems.

The system tracks changes over time and can identify developing abnormalities.

This supports a combination of: **scheduled maintenance** and **condition-based maintenance**.

The common architecture of the AH-71 and UH-71 also allows maintenance organizations to share many components, tools, diagnostic equipment, and training requirements.

19. COMMON AIRCRAFT FAMILY

Although the AH-71 and UH-71 perform different missions, personnel should recognize the common systems between them.

Both aircraft share the same fundamental:

- propulsion philosophy;
- rotor architecture;
- transmission architecture;
- flight-control architecture;
- avionics philosophy;
- maintenance systems;
- and digital infrastructure.

For pilots transitioning between variants, this commonality provides a familiar foundation.

For ground personnel, it means that the visual appearance and operating behavior of the two aircraft may differ according to configuration, while their underlying systems remain closely related.

20. FUTURE DEVELOPMENT

The H-71 has been designed with future expansion in mind.

The aircraft architecture provides room for additional:

- sensors;
- communications systems;
- electronic warfare equipment;
- navigation technology;
- mission computers;
- autonomous assistance;
- and other mission equipment.

The objective is to allow the aircraft family to evolve without requiring a complete redesign of the underlying platform.

21. SUMMARY

The H-71 represents a common rotary-wing platform developed for two distinct purposes.

The **AH-71** provides the UHR with an attack and reconnaissance aircraft built around sensors, maneuverability, weapons integration, and battlefield observation.

The **UH-71** provides a utility and assault aircraft capable of transporting personnel, equipment, casualties, and specialized mission loads.

Both aircraft share a common technological foundation developed through the combined efforts of **Prelancer Technologies** and **ELTRON Aviation**.

For pilots, the H-71 provides a modern flight-control and avionics environment designed around aircraft control, information availability, and mission flexibility.

For ground units, it provides a recognizable and adaptable aviation platform capable of supporting forces across a wide range of operations.

The H-71 is ultimately intended to function as more than a helicopter.

It is a **common aviation system connecting pilots, aircraft, and ground forces into the same operational environment**.

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